



KOLEJ MATRIKULASI NEGERI SEMBILAN
UJIAN SELARAS 2
DP024 FIZIK (2 JAM)

NAMA : _____

PRAKTIKUM : _____

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

<i>Untuk Kegunaan Pemeriksa</i>		
NO. SOALAN	MARKAH PENUH	MARKAH DIPEROLEH
1	16	
2	12	
3	10	
JUMLAH	38	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|---|
| <p>1. $x = A \sin \omega t$</p> <p>2. $\omega = \frac{2\pi}{T} = 2\pi f$</p> <p>3. $v = A\omega \cos \omega t$</p> <p>4. $a = -A\omega^2 \sin \omega t$</p> <p>5. $k = \frac{2\pi}{\lambda}$</p> <p>6. $v = f\lambda$</p> <p>7. $y(x, t) = A \sin(\omega t \pm kx)$</p> <p>8. $v = \frac{dy}{dt}$</p> <p>9. $y = 2A \cos kx \sin \omega t$</p> <p>10. $v = \sqrt{\frac{T}{\mu}}$</p> <p>11. $I = \frac{dQ}{dt}$</p> | <p>12. $Q = ne$</p> <p>13. $\rho = \frac{RA}{l}$</p> <p>14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$</p> <p>15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$</p> <p>16. $V = IR$</p> <p>17. $C = \frac{Q}{V}$</p> <p>18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$</p> <p>19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$</p> <p>20. $\tau = RC$</p> <p>21. $Q = Q_o e^{-t/RC}$</p> <p>22. $Q = Q_o (1 - e^{-t/RC})$</p> |
|--|---|

- 1 (a) The progressive wave equation is given as

$$y(x, t) = 5.0 \sin(12.6t + 1.6x)$$

where y and x are in centimeters (cm) and t is in seconds (s). Determine the

- (i) amplitude of the wave.
- (ii) frequency of the wave.
- (iii) periodic time.
- (iv) wavelength of the wave.
- (v) speed of the wave.
- (vi) displacement of the wave at $t = 4$ s and $x = 20$ cm.

[9 marks]

- (b) Two progressive waves travelling in opposite direction form a standing wave with wave function given by

$$y = 0.03 \cos \frac{x}{2} \sin 40t$$

Where y and x are in meter and t in second.

Write the equations of both progressive waves that form the standing wave.

[4 marks]

- (c) Consider a string in a guitar whose length is 80 cm and a mass of 0.32 g with tension 80 N is plucked. Compute the velocity of wave produced by the guitar string.

[3 marks]

- 2 (a) (i) The amount of charge flows through a cross-sectional area of the copper wire is 538 C. Determine the current flows in the copper wire in 95 s.
- (ii) Current of 0.8 A flows in a light bulb. Calculate the number of electrons that enter the light bulb every minute.

[4 marks]

- (b) A 2.0 m length conducting wire with cross-sectional area of 0.785 mm^2 and 3.0Ω resistance. Calculate

- (i) the resistivity of the wire.
- (ii) the potential difference across the wire if current flow is 1.5 A.

[4 marks]

- (c)

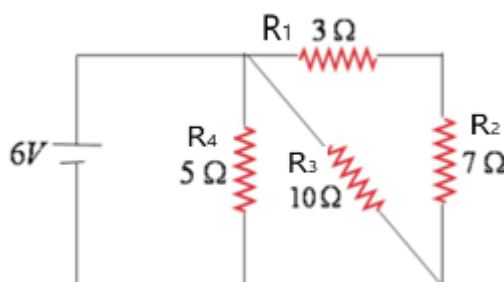


FIGURE 1

FIGURE 1 shows three resistors are connected to a supply voltage, V . Determine

- the effective resistance.
- potential difference across resistor, R_3 .

[4 marks]

3 (a)

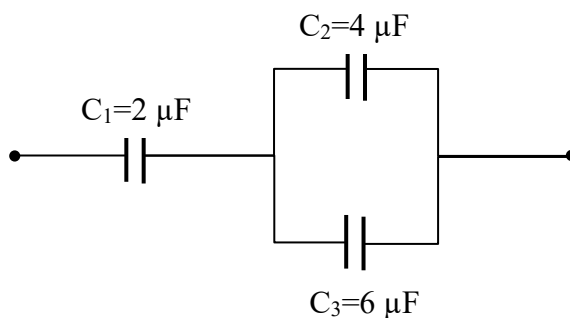


FIGURE 2

FIGURE 2 shows three capacitors are connected to 10 V supply. Calculate

- the effective capacitance.
- the charge in capacitor C_1 .

[4 marks]

- (b) A capacitor of capacitance $2 \mu\text{F}$ is charged using a battery of e.m.f 1.5 V through a $60 \text{ k}\Omega$ resistor. Determine

- time constant of the charged circuit.
- time taken for the charge on the capacitor to increase to 40 % of its maximum value.
- the charge in the capacitor after 0.10 s.

[6 marks]

4 (a)

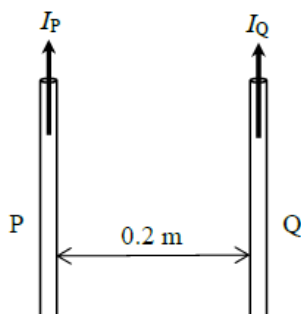


FIGURE 4.1

FIGURE 4.1 shows two long straight parallel wires, P and Q are placed 0.2 m apart. Wire P carries a current of 2.5 A and wire Q carries a current of 3.0 A. If the direction of current for both wires are upward, determine the magnitude **and** direction of the resultant magnetic field at midpoint between the wires.

[5 marks]

- (b) (i) A circular coil has 28 turns and a radius of 24 cm. If the current flowing in the coil is 5.5 A, calculate the magnetic field strength at the centre of the coil.
- (ii) A magnetic field of strength 0.9 T is directed perpendicular to the plane of a circular coil of area 2.5 cm^2 . Calculate the magnetic flux through the area enclosed by this circular coil.
- (iii) Calculate the flux linkage in a circular coil of 25 turns and area 30 cm^2 in a magnetic field of strength 7 T parallel to the plane.

[7 marks]

(c)

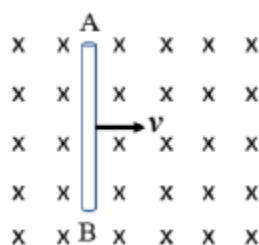


FIGURE 4.2 shows a 1.2 m conducting rod AB is placed in a uniform magnetic field with magnitude of 4.0 T. Then, the rod is pulled to the right with a uniform velocity of 3.5 m s^{-1} perpendicular to magnetic field. Determine the magnitude of the induced emf in the rod.

[2 marks]

- 5 (a) An object of height 22.0 cm is placed 30.0 cm in front of a concave mirror. If the height of image is 58.0 cm and virtual image is formed,
- (i) sketch and label a ray diagram to show the formation of the image.
- (ii) calculate the focal length of the mirror.
- (b) An object is located 20.0 cm to the left of a diverging lens having a focal length of 32.0 cm.
- (i) Determine the location of the image.
- (ii) Calculate the magnification of the image.
- (iii) Construct a ray diagram for this arrangement.

[7 marks]

- 6 (a) In a double-slit experiment, the slit separation is 0.03 cm, and the distance of the double-slit from the screen is 160 cm. When a green light is used, the distance of the first bright fringe from the central maximum is 0.3 cm. Calculate the
- (i) wavelength of the green light.
 - (ii) distance of the third dark fringes from the central maximum.

[6 marks]

- (b) Two slits with a separation of 0.05 mm are illuminated by light at 620 nm, and the interference pattern is observed on screen 3.00 m from the slits. Calculate the
- (i) distance of second dark fringe from central bright.
 - (ii) distance between the second dark fringe and third bright fringe.
 - (iii) fringes separation.

[8 marks]